



Capstone Project: Comprehensive SOC Incident Response

1. Introduction

This capstone project demonstrates a complete end-to-end SOC workflow, integrating attack simulation, adversary emulation, detection, triage, response, automation, forensic analysis, metrics evaluation, and executive reporting.

The simulation environment utilized:

- **Rapid7 Metasploit**
- **MITRE Corporation Caldera**
- **Wazuh Inc. Wazuh**
- **CrowdSec CrowdSec**
- **StrangeBee TheHive**
- **Elastic Security**

The exercise validates SOC readiness against exploitation and lateral movement attempts.

2. Attack Simulation

2.1 Exploitation Using Metasploit

A vulnerable Metasploitable2 machine was targeted using:

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exploit/multi/samba/usermap_script
```

This exploited a Samba vulnerability to gain remote shell access.

MITRE Mapping:

T1210 – Exploitation of Remote Services



3. Adversary Emulation

Using MITRE Caldera:

- Simulated lateral movement behavior.
- Executed scripted command activity.
- Tested detection controls.

Detection Log

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 16:00	192.168.1.102	Samba Exploit Detected	T1210

Wazuh successfully generated alerts within minutes.

4. Detection & Triage

4.1 Alert Review

- Verified source IP.
- Confirmed exploit signature.
- Classified severity as Critical.
- Checked correlated logs.

Alert confirmed as True Positive.

4.2 Case Creation in TheHive

A case was created with:

- Exploit details
- Source IP
- Affected system
- Evidence artifacts

Severity: Critical

Status: Under Investigation

5. Response & Containment

Immediate containment actions:

- Isolated compromised VM.
- Blocked attacker IP using CrowdSec.
- Verified block using ping test (failed connection).
- Terminated malicious sessions.

Containment was successful.



6. SOAR Automation

Automation workflow implemented:

- Extract IP from alert.
- Query reputation.
- If malicious → Block via CrowdSec.
- Create TheHive case.
- Notify SOC.

Automation execution confirmed successful.

7. Post-Incident Root Cause Analysis

7.1 5 Whys

Question	Answer
Why was system compromised?	Samba vulnerability exploited
Why vulnerable?	Patch not applied
Why not patched?	Update schedule misconfigured
Why misconfigured?	Lack of periodic review
Why no review?	Weak governance oversight

Root Cause: Failure in patch management process.

7.2 Fishbone Causes

- Technology: Outdated software
 - Process: Weak patch management
 - People: No vulnerability awareness review
 - Policy: No compliance enforcement
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8. Security Metrics Calculation

Metric	Value
MTTD	10 minutes
MTTR	45 minutes
Dwell Time	55 minutes

Analysis: Rapid detection and containment reduced dwell time significantly.



9. Incident Report (SANS Format)

Executive Summary

A controlled exploitation of a Samba vulnerability was conducted to validate SOC detection and response capabilities. The attack was executed using Metasploit and detected by Wazuh. The alert was classified as critical and escalated to the incident response team. Immediate containment actions were implemented, including system isolation and IP blocking via CrowdSec. No lateral movement beyond the test system was observed.

Timeline

The exploit was initiated at 16:00. Alert generation occurred within minutes. SOC triage confirmed malicious exploitation. The compromised system was isolated, and the attacker IP was blocked within 45 minutes of detection. Post-incident forensic analysis validated no persistence mechanisms remained active.

Root Cause Analysis

The incident occurred due to an unpatched Samba service vulnerability. Weak patch governance and lack of configuration audits contributed to the exposure.

Recommendations

Implement automated patch management, continuous vulnerability scanning, stricter firewall segmentation, and regular adversary emulation exercises.

10. Executive Briefing

A controlled cybersecurity simulation identified a vulnerability in a server running outdated Samba software. The issue was exploited in a test environment to evaluate detection and response capabilities. The SOC successfully detected the exploit within minutes and contained the incident in under one hour. Immediate actions included isolating the affected system and blocking the attacker's access. No data loss or lateral movement occurred. The root cause was identified as delayed patch management procedures. Improvements including automated patch deployment, continuous vulnerability monitoring, and enhanced governance oversight are recommended. Overall, the SOC demonstrated strong detection and response capability, with opportunities to strengthen preventive controls.



11. Key Outcomes

- Exploit detected quickly.
 - Automation reduced response time.
 - Forensic analysis confirmed scope.
 - Governance gaps identified.
 - Metrics validated SOC efficiency.
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12. Conclusion

This comprehensive capstone successfully demonstrated an advanced SOC workflow integrating detection, automation, emulation, forensic analysis, root cause investigation, and executive reporting. The exercise validated monitoring effectiveness while highlighting process improvements. Continuous adversary simulation and metrics-driven analysis strengthen long-term organizational cybersecurity resilience.